

Mohit Nalavadi

Senior Software Engineer (Data, ML)

Contact

- +49 174 5423 450
- mnalavadi@gmail.com
- github.com/momonala
- linkedin.com/in/mohit-nalavadi/

Education

- **University of Twente** 2018
  M.S. Biometrics & Data Science
- **Santa Clara University** 2016
  B.Eng. Biomolecular Engineering

Skills

Software:

- Python, SQL, Bash, Arduino, Git
- GCP, PostgreSQL, BigQuery, MongoDB
- Docker, Kubernetes
- ETL pipelines
- Apache Beam, Airflow, Kafka
- Cursor, Claude Code, Codex
- Scientific computing, data visualization
- Ansible, MLOps
- Monitoring, logging, Datadog
- REST APIs, Linux
- CI/CD, unit/end-to-end testing

Other:

- Agile, technical writing, scientific communication, project management
- Remote sensing
- Hardware/sensors
- Biology/chemistry

Awards

- Berlin Hack and Tell Winner Jan. 2019
- Young Scientist Program Graduate – Blue Marble Space Institute of Science @NASA 2017
- Best Interdisciplinary Thesis – Santa Clara University 2016

About

ML/Data Engineer with 8+ years of experience building scalable, data intensive systems on GCP, working with large-scale datasets and real-time applications across digital pathology, remote sensing, and e-commerce

Experience

Senior ML Engineer @ AIgnostics GmbH **2025-2026**

- Worked on ML systems for digital pathology foundation models (~2B parameters, 5M+ training images) for cancer research
- Worked with product team to refine our metrics pipeline for GA release
- Migrated our Docker builder from Kaniko to BuildKit and cut CI build times by 50%
- Optimized our inference engine from TensorFlow to TensorRT
- Built compliance tooling for traceability of our tests and deployments

Senior Data Engineer @ Grandperspective GmbH **2021-2025**

- Engineered data infrastructure for real-time, 24/7 remote sensing (gas leaks) detecting 400+ gases across km-scale sites in seconds.
- Helped grow the client base by 5x
- Built ETL pipelines and a backup system for 4TB+ of sensor data (24GB/day per system)
- Built a data visulization system for high dimensional FTIR images
- Worked with Data Science and DevOps teams to ensure 99% platform uptime & data availability.

ML Engineer @ Commercetools GmbH **2018-2021**

- Built and scaled ML APIs for high-traffic e-commerce applications
- Developed an image similarity search engine over 500k+ images, returning nearest neighbors in <2s for real-time product discovery
- Designed and operated a MongoDB index recommendation system analyzing weekly production logs, generating 10–30 high-impact indexes. Improve query performance up to 50%
- Created ML services for demand forecasting and customer segmentation, enabling data-driven recommendations

Freelancer/Intern **2016-2017**

Amyris: Automation Engineer Intern

- Wrote software for robotic automation in synthetic biology
- Improved computer vision algorithms, increasing accuracy to 95%+

General Assembly: Data Science Associate Instructor

- Taught ML and data visualization in a 10-week course to 20 students

Leif Therapeutics – Hardware QA Engineer

- Built and tested 60 beta units of a wearable ECG (haptic biofeedback)
- Resolved system-level issues on pre-production hardware and firmware

NASA Ames: Biomechanics Working Student **2014-2017**

- Studied skeletal health in microgravity across 10+ studies
- Wrote algorithms to process data for mechanical strength tests
- µCT 3D imaged (µCT) 150+ specimen, conducted material strength tests